

Muhammad Kashif, Ph.D.

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Citation metrics: 1,150+ citations; *h-index* 21; *i10-index* 33 (October 2026)

Research Profile

My research focuses on quantum machine learning and hybrid quantum-classical computing, with the goal of making quantum learning methods more reliable, understandable, and resource-efficient. I study how model architecture, data encoding, learning objectives, and noise influence trainability and performance. I also develop resource-aware design and evaluation methods to assess the computational costs and practical benefits of hybrid models. My work extends to AI-assisted quantum programming, including domain-specific language models, datasets, and benchmarks for quantum code generation. My background also includes FPGA-based cryptographic accelerators and model-based embedded system verification. This experience connects my current quantum computing research with system design and design automation.

Research interests: Quantum computing; quantum machine learning; hardware-aware quantum computing; hybrid quantum-classical system design; quantum resource estimation; quantum design automation and resource-aware neural architecture search; trainability and noise robustness; AI-assisted quantum programming.

Professional Experience

Postdoctoral Research Associate

Aug 2023–Present

[Center for Quantum and Topological Systems](#), New York University Abu Dhabi, UAE

Additional current role: Research Team Lead, [eBRAIN Lab](#); coordinate collaborative research and mentor undergraduate and graduate researchers.

- Develop and evaluate hybrid quantum-classical learning architectures, with a focus on trainability, gradient propagation, and robustness under noise.
- Investigate how data encoding, circuit architecture, learning objectives, and observable selection influence quantum neural network performance.
- Develop analytical and hardware-calibrated quantum resource cost models and resource-aware neural architecture search methods to evaluate accuracy–cost trade-offs.
- Assess the computational benefits and limitations of hybrid quantum-classical models through comparative benchmarking and resource analysis.
- Contribute to domain-specific language models, datasets, and benchmarks for automated quantum code generation using PennyLane.
- Investigate quantum optimization and hybrid workflows for financial applications, path planning, unit commitment, and robot navigation.
- Coordinate collaborative projects and supervise student research in quantum machine learning, optimization, and quantum software.

Research and Teaching Assistant

Mar 2018–Dec 2019

Istanbul Şehir University, Turkey

- Conducted undergraduate laboratory and physics problem-solving sessions for 60+ students.
- Designed, optimized, and evaluated FPGA-based elliptic curve cryptographic accelerators.

Research Engineer (Remote)

Feb 2015–Aug 2017

Collaborative research projects, National Science Technology and Innovation Plan (NSTIP), Saudi Arabia

- Developed structural and behavioral SysML/MARTE models of safety-critical embedded systems, including unmanned aerial vehicles, intelligent traffic controllers, and collision avoidance systems.
- Implemented FPGA-based elliptic curve crypto-processors and optimized area and throughput.
- Implemented sorting algorithms in C for MATLAB-based genetic algorithm workflows supporting temporal verification and execution-time prediction.

Education

- Ph.D., Computer Science and Engineering** 2020–2023
Hamad Bin Khalifa University, Qatar
Thesis: *Quantum Machine Learning on Noisy Intermediate-Scale Quantum Devices.*
- M.Sc., Electronics and Computer Engineering** 2018–2019
Istanbul Şehir University, Turkey
Thesis: *A Hardware-Efficient Elliptic Curve Accelerator for FPGA-Based Cryptographic Applications.*
- B.Sc., Electrical (Electronics) Engineering** 2011–2015
COMSATS, Pakistan
Final-year project: *Home Automation System Using Raspberry Pi over Wi-Fi.*

Teaching, Supervision, and Mentoring

Teaching and Educational Materials

- **Undergraduate teaching assistance, Istanbul Şehir University (2018–2019):** Conducted undergraduate laboratory instruction and physics problem-solving sessions for more than 60 students.
- **Educational materials development, NSTIP research projects, Saudi Arabia (2015–2017):** Prepared tutorials and case studies in model-based systems engineering to support course development.

Thesis Co-Supervision

- **M.Sc., Université Libre de Bruxelles (2024–2025):** *Noise Mitigation Techniques for Reliable Quantum Neural Networks.*
- **B.Sc., New York University Abu Dhabi (2023–2024):** *Comparative Analysis of Noise and Robustness Evaluation in Quantum Neural Networks.*

Research and Hackathon Mentoring

- Supervise undergraduate and graduate research projects at NYU Abu Dhabi in quantum machine learning, optimization, and quantum software.
- **Mentor, NYUAD Quantum Hackathon for Social Good, 2024, 2025, and 2026.**
- **Mentor, Quantum Computing Hackathon Pakistan, 2026;** mentored team placed second runner-up.

Teaching Interests

Quantum computing; quantum machine learning; machine learning and AI; digital logic and systems design; computer architecture; embedded systems; FPGA-based hardware design.

Awards and Honors

- **Best Paper Nominee, IJCNN 2026:** [Design Space Exploration of Hybrid Quantum Neural Networks for Chronic Kidney Disease.](#)
- **Best Paper Nominee, IJCNN 2026:** [A PennyLane-Centric Dataset to Enhance LLM-Based Quantum Code Generation Using RAG.](#)
- **First Prize, QInnovation World Challenge,** Thales, Quantum Innovation Summit, Dubai, UAE, 2025. [Challenge](#)
- **Best Researcher Award,** eBRAIN Lab, NYU Abu Dhabi, 2024–2025.
- **Fully funded Ph.D. scholarship,** Hamad Bin Khalifa University, 2020–2023.
- **Fully funded M.Sc. scholarship (TA/RA),** Istanbul Şehir University, 2018–2019.
- **Outstanding Performance Award,** NSTIP, Saudi Arabia, 2015–2017.

Professional Service and Leadership

- **Research leadership:** Research Team Lead, eBRAIN Lab, NYU Abu Dhabi; coordinate collaborative research and student mentoring.
- **Technical program committee:** Design, Automation & Test in Europe (DATE), 2027, [D16 track: Design Automation for Quantum Computing](#); [IEEE International Workshop on Quantum Computing: Circuits, Systems, Automation, and Applications \(QC-CSAA\)](#), co-located with ISVLSI, 2026.

- **Workshop and special-session organization:** Co-organized workshop on [Applied Quantum Machine Learning: From Quantum Circuits to Smart Systems](#) at IEEE IJCNN; Co-organized [QGenAI: Synergies between Quantum Computing and Generative Artificial Intelligence](#) at 2026 IEEE Quantum Week (QCE); Co-organized special session on [Hardware-Software Co-Design of Quantum Machine Learning Systems](#) in IEEE Embedded Systems Week 2026.
- **Challenge co-organization:** [Quantum Federated Learning for Fraud Detection](#), Quantum Innovation Summit, Dubai, 2025.
- **Journal reviewing:** IEEE Transactions on Emerging Topics in Computing; IEEE Transactions on Quantum Engineering; IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems; Advanced Quantum Technologies; Engineering Applications of Artificial Intelligence; Computers & Electrical Engineering; Expert Systems with Applications; Journal of Computational Physics; Neurocomputing; SoftwareX.
- **Conference reviewing:** IEEE International Conference on Quantum Computing and Engineering (QCE); International Joint Conference on Neural Networks (IJCNN); Design Automation and Test in Europe (DATE)

Selected Conference Presentations

- **DAC 2026**, Long Beach, California, USA, July 2026: *Closing the Loop: Resource-Aware Hybrid NAS Guided by Analytical and Hardware-Calibrated Quantum Cost Modeling.*
- **DAC 2025**, San Francisco, California, USA, June 2025: *Computational Advantage in Hybrid Quantum Neural Networks: Myth or Reality?*
- **IEEE Quantum Week (QCE) 2025**, New Mexico, USA, September 2025: *PennyCoder: Efficient Domain-Specific LLMs for PennyLane-Based Quantum Code Generation*
- **ICRC 2024**, San Diego, California, USA, December 2024: *The Dilemma of Random Parameter Initialization and Barren Plateaus in Variational Quantum Algorithms.*

Technical Skills

- **Quantum computing:** PennyLane, Qiskit, IBM and IQM Quantum backends; variational quantum algorithms; hybrid quantum neural networks; noise modeling; quantum resource estimation; robustness and trainability analysis.
- **Machine learning and scientific computing:** Python, NumPy, SciPy, scikit-learn, Keras, TensorFlow; PyTorch; experimental benchmarking.
- **Optimization and system evaluation:** Resource-aware neural architecture search; analytical and hardware-calibrated cost modeling; circuit-depth and two-qubit gate analysis; accuracy–cost trade-off evaluation.
- **Programming and hardware design:** C/C++, MATLAB, Verilog, VHDL, FPGA, Vivado, Xilinx.
- **System modeling and verification:** SysML, MARTE, model-based embedded system design and verification.

Selected Research Contributions

Trainability and robustness of quantum learning models: Investigated how residual connections, circuit design, learning objectives, and noise affect optimization and gradient propagation in quantum neural networks. Representative publications include [ResQNets](#) (EPJ Quantum Technology, 2024); [Deep Quantvolutional Neural Networks with Enhanced Trainability and Gradient Propagation](#) (Scientific Reports, 2025), [The Impact of Cost Function Globality and Locality in HQNNs](#) (Machine Learning: Science and Technology, 2023); and [Alleviating Barren Plateaus in Parameterized Quantum Machine Learning Circuits](#) (DATE, 2024).

Resource-aware hybrid quantum–classical computing: Studied the computational costs and potential benefits of hybrid quantum neural networks and developed resource-aware architecture search methods. Representative publications include [Computational Advantage in Hybrid Quantum Neural Networks: Myth or Reality?](#) (DAC, 2025); [FAQNAS](#) (DATE, 2026); and [Closing the Loop: Resource-Aware Hybrid NAS Guided by Analytical and Hardware-Calibrated Quantum Cost Modeling](#) (DAC, 2026).

AI-assisted quantum programming: Contributed to language models, datasets, and evaluation benchmarks for PennyLane-based quantum code generation. Representative publications include [PennyCoder](#) (QCE, 2025); [QHackBench](#) (QAI, 2025); and [A PennyLane-Centric Dataset to Enhance LLM-Based Quantum Code Generation Using RAG](#) (IJCNN, 2026).

Journal Articles

- J1 **M. Kashif** and S. Al-Kuwari. “ResQNets: A residual approach for mitigating barren plateaus in quantum neural networks.” *EPJ Quantum Technology*, 2024. DOI

- J2 **M. Kashif** and M. Shafique. “Deep quantum convolutional neural networks with enhanced trainability and gradient propagation.” *Scientific Reports*, 2025. [DOI](#)
- J3 **M. Kashif** and S. Al-Kuwari. “The impact of cost function globality and locality in hybrid quantum neural networks on NISQ devices.” *Machine Learning: Science and Technology*, 2023. [DOI](#)
- J4 T. Ahmed, **M. Kashif**, A. Marchisio, and M. Shafique. “A comparative analysis and noise robustness evaluation in quantum neural networks.” *Scientific Reports*, 2025. [DOI](#)
- J5 **M. Kashif** and S. Al-Kuwari. “Physical realization of measurement based quantum computation.” *IEEE Access*, 2023. [Publisher](#)
- J6 **M. Kashif** and S. Al-Kuwari. “The unified effect of data encoding, ansatz expressibility and entanglement on the trainability of HQNNs.” *International Journal of Parallel, Emergent and Distributed Systems*, 2023. [DOI](#)
- J7 M. Rashid, M. Imran, **M. Kashif**, and A. Sajid. “An optimized architecture for binary Huff curves with improved security.” *IEEE Access*, 2021. [DOI](#)
- J8 **M. Kashif** and I. Çiçek. “Design space exploration of hybrid quantum–classical neural networks.” *Electronics*, 10(23), 2980, 2021. [Publisher](#)
- J9 **M. Kashif** and I. Çiçek. “Field-programmable gate array (FPGA) hardware design and implementation of a new area efficient elliptic curve crypto-processor.” *Turkish Journal of Electrical Engineering and Computer Sciences*, 29(4), 2127–2139, 2021. [DOI](#)
- J10 M. W. Anwar, M. Rashid, F. Azam, A. Naeem, **M. Kashif**, and W. H. Butt. “A unified model-based framework for the simplified execution of static and dynamic assertion-based verification.” *IEEE Access*, 2020. [DOI](#)
- J11 M. Rashid, S. A. B. Shah, M. Arif, and **M. Kashif**. “Determination of worst-case data using an adaptive surrogate model for real-time system.” *Journal of Circuits, Systems and Computers*, 2020. [DOI](#)
- J12 M. W. Anwar, M. Rashid, F. Azam, **M. Kashif**, and W. H. Butt. “A model-driven framework for design and verification of embedded systems through SystemVerilog.” *Design Automation for Embedded Systems*, 2019. [DOI](#)
- J13 M. Imran, M. Rashid, A. R. Jafri, and **M. Kashif**. “Throughput/area optimised pipelined architecture for elliptic curve crypto processor.” *IET Computers & Digital Techniques*, 2019. [DOI](#)
- J14 M. W. Anwar, M. Rashid, F. Azam, and **M. Kashif**. “Model-based design verification for embedded systems through SVOCL: An OCL extension for SystemVerilog.” *Design Automation for Embedded Systems*, 2017. [DOI](#)

Published Conference and Proceedings Papers

- C1 **M. Kashif**, A. Marchisio, and M. Shafique. “Computational advantage in hybrid quantum neural networks: Myth or Reality?” *ACM/IEEE Design Automation Conference (DAC)*, 2025. [DOI](#)
- C2 **M. Kashif**, S. Khalid, A. Marchisio, N. Innan, and M. Shafique. “FAQNAS: FLOPs-aware hybrid quantum neural architecture search using genetic algorithm.” *Design, Automation & Test in Europe (DATE)*, 2026. [DOI](#)
- C3 **M. Kashif** and M. Shafique. “Position Paper: Quantum neural networks—A paradigm shift in AI or a theoretical promise?” *International Joint Conference on Neural Networks (IJCNN)*, 2025. [DOI](#)
- C4 **M. Kashif**, S. Khalid, N. Innan, A. Marchisio, and M. Shafique. “Evaluating quantum amplitude estimation for pricing multi-asset basket options.” *IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 2025. [DOI](#)
- C5 **M. Kashif** and M. Shafique. “HQNET: Harnessing quantum noise for effective training of quantum neural networks in NISQ era.” *IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 2025. [DOI](#)
- C6 **M. Kashif**, et al. “Alleviating barren plateaus in parameterized quantum machine learning circuits: Investigating advanced parameter initialization strategies.” *Design, Automation & Test in Europe (DATE)*, 2024. [DOI](#)
- C7 **M. Kashif**, E. Sychiuco, and M. Shafique. “Investigating the effect of noise on the training performance of hybrid quantum neural networks.” *International Joint Conference on Neural Networks (IJCNN)*, 2024. [DOI](#)
- C8 **M. Kashif** and M. Shafique. “NRQNN: The role of observable selection in noise-resilient quantum neural networks.” *CSCE 2024, revised selected papers, Communications in Computer and Information Science*, vol. 2257, Springer, 2025 (conference held 2024). [DOI](#)
- C9 **M. Kashif** and M. Shafique. “The dilemma of random parameter initialization and barren plateaus in variational quantum algorithms.” *IEEE International Conference on Rebooting Computing (ICRC)*, 2024. [DOI](#)

- C10 **M. Kashif** and S. Al-Kuwari. “Demonstrating quantum advantage in hybrid quantum neural networks for model capacity.” *IEEE International Conference on Rebooting Computing (ICRC)*, 2022. DOI
- C11 **M. Kashif** and S. Al-Kuwari. “Qiskit as a simulation platform for measurement-based quantum computation.” *IEEE International Conference on Software Architecture Companion (ICSA-C)*, 2022. DOI
- C12 A. Marchisio, E. Sychiuco, **M. Kashif**, and M. Shafique. “Cutting is all you need: Execution of large-scale quantum neural networks on limited-qubit devices.” *IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 2025.
- C13 N. Innan, **M. Kashif**, et al. “Next-generation quantum neural networks: Enhancing efficiency, security, and privacy.” *IEEE International Symposium on On-Line Testing and Robust System Design (IOLTS)*, 2025. Publisher
- C14 T. Ahmed, A. Marchisio, **M. Kashif**, and M. Shafique. “Noisy HQNNs: A comprehensive analysis of noise robustness in hybrid quantum neural networks.” *International Joint Conference on Neural Networks (IJCNN)*, 2025. DOI
- C15 A. Basit, M. Shao, M. H. Asif, N. Innan, **M. Kashif**, et al. “PennyCoder: Efficient domain-specific LLMs for PennyLane-based quantum code generation.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2025. DOI
- C16 A. Marchisio, M. U. Hafeez, N. Innan, **M. Kashif**, and M. Shafique. “Q-PORT: Quantum portfolio optimization with resource-efficient encoding and scalability analysis.” *International Conference on Quantum Engineering Sciences and Technologies for Industry and Services (QUEST-IS)*, 2025. DOI
- C17 N. Innan, **M. Kashif**, A. Marchisio, Y. S. Gan, F. Barbaresco, and M. Shafique. “QUAV: Quantum-assisted path planning and optimization for UAV navigation with obstacle avoidance.” *IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 2025. DOI
- C18 A. Basit, M. Shao, M. H. Asif, N. Innan, **M. Kashif**, et al. “QHackBench: Benchmarking large language models for quantum code generation using PennyLane hackathon challenges.” *IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 2025. DOI
- C19 N. Innan, **M. Kashif**, et al. “Quantum amplitude estimation in practice: A case study in option pricing.” *International Conference on Quantum Engineering Sciences and Technologies for Industry and Services (QUEST-IS)*, 2025. DOI
- C20 J. Kaldari, **M. Kashif**, S. Al-Kuwari, and M. Shafique. “ResQGRNN: Quantum-compatible residual learning for graph recurrent neural networks.” *IEEE International Conference on Quantum Artificial Intelligence (QAI)*, 2025. DOI
- C21 K. Zaman, T. Ahmed, **M. Kashif**, et al. “Studying the impact of quantum-specific hyperparameters on hybrid quantum–classical neural networks.” *CSCE 2024, revised selected papers, Communications in Computer and Information Science, vol. 2257, Springer*, 2025 (conference held 2024). DOI
- C22 K. Zaman, A. Marchisio, **M. Kashif**, and M. Shafique. “PO-QA: A framework for portfolio optimization using quantum algorithms.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2024. DOI
- C23 M. Zahoor, F. Azam, M. W. Anwar, N. Yousaf, and **M. Kashif**. “A UML profile for the service discovery in the Enterprise Cloud Bus (ECB) framework.” *Complex, Intelligent, and Software Intensive Systems (CISIS)*, Springer, 2019. DOI
- C24 **M. Kashif**, I. Çiçek, and M. Imran. “A hardware efficient elliptic curve accelerator for FPGA based cryptographic applications.” *International Conference on Electrical and Electronics Engineering (ELECO)*, 2019. DOI
- C25 M. Rashid, M. W. Anwar, F. Azam, and **M. Kashif**. “Model-based requirements and properties specifications trends for early design verification of embedded systems.” *System of Systems Engineering Conference (SoSE)*, 2016. DOI
- C26 M. Imran, **M. Kashif**, and M. Rashid. “Hardware design and implementation of scalar multiplication in elliptic curve cryptography (ECC) over GF (2^{163}) on FPGA.” *International Conference on Information and Communication Technologies (ICICT)*, 2015. DOI

Accepted Conference Papers — Manuscript Links

- A1 **M. Kashif**, A. Marchisio, and M. Shafique. “Closing the Loop: Resource-aware Hybrid NAS Guided by Analytical and Hardware-Calibrated Quantum Cost Modeling.” *ACM/IEEE Design Automation Conference (DAC)*, 2026; accepted and presented; proceedings pending. [arXiv:2603.00625](https://arxiv.org/abs/2603.00625)

- A2 **M. Kashif** and M. Shafique. “Late Breaking Results: Hardware-aware compilation reshapes trainability in variational quantum circuits.” *ACM/IEEE Design Automation Conference (DAC)*, 2026; accepted and presented; proceedings pending. [arXiv:2604.16527](#)
- A3 **M. Kashif**, H. M. Siraj, N. Innan, A. Marchisio, and M. Shafique. “Design space exploration of hybrid quantum neural networks for chronic kidney disease.” *International Joint Conference on Neural Networks (IJCNN)*, 2026; accepted and presented; *Best Paper Nominee*. [arXiv:2604.13608](#)
- A4 A. Basit, N. Innan, M. H. Asif, M. Shao, **M. Kashif**, A. Marchisio, and M. Shafique. “A PennyLane-Centric Dataset to Enhance LLM-based Quantum Code Generation using RAG.” *International Joint Conference on Neural Networks (IJCNN)*, 2026; accepted and presented; *Best Paper Nominee*. [arXiv:2503.02497](#)
- A5 T. Ahmed, A. Marchisio, **M. Kashif**, N. Innan, and M. Shafique. “GAT-QNN: Genetic algorithm-based training of hybrid quantum neural networks.” *International Joint Conference on Neural Networks (IJCNN)*, 2026; accepted and presented. [arXiv:2604.15048](#)
- A6 M. Shao, N. Innan, H. Janardhanan, **M. Kashif**, A. Marchisio, and M. Shafique. “PennySynth: RAG-driven data synthesis for automated quantum code generation.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026; accepted and presented. [arXiv:2605.25572](#)
- A7 M. K. Altrabulsi, N. Innan, A. Marchisio, **M. Kashif**, and M. Shafique. “Q-SpiRL: Quantum spiking reinforcement learning for adaptive robot navigation.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2026; accepted and presented. [arXiv:2605.20801](#)
- A8 A. Marchisio, A. Ebrahim, N. Innan, **M. Kashif**, and M. Shafique. “QLIF-CAST: Quantum leaky-integrate-and-fire for time-series weather forecasting.” *IEEE International Quantum Artificial Intelligence (QAI)*, 2026; accepted. [arXiv:2605.18333](#)
- A9 J. R. M. Njiki, N. Innan, A. Marchisio, **M. Kashif**, J. M. Dricot, and M. Shafique. “Robustness evaluation of hybrid quantum neural networks under noise models via system-level error mitigation.” *IEEE International Quantum Artificial Intelligence (QAI)*, 2026; accepted. [arXiv:2604.17515](#)

Preprints

- P1 **M. Kashif** and M. Shafique. “Beyond Logical Circuits: Hardware-aware analysis of expressibility and trainability in variational quantum algorithms.” *arXiv preprint*, 2026. [arXiv:2605.25552](#)
- P2 **M. Kashif** and M. Shafique. “Rethinking expressibility–trainability trade-off in hybrid quantum neural networks.” *arXiv preprint*, 2026. [arXiv:2605.25768](#)
- P3 A. Marchisio, **M. Kashif**, N. Innan, and M. Shafique. “Hybrid quantum–classical neural architecture search.” *arXiv preprint*, 2026. [arXiv:2605.18345](#)
- P4 R. Ahmed, **M. Kashif**, N. Innan, and M. Shafique. “Quantum vs. classical machine learning: A benchmark study for financial prediction.” *arXiv preprint*, 2026. [arXiv:2601.03802](#)
- P5 M. Atallah, N. Innan, **M. Kashif**, and M. Shafique. “Investigating different barren plateaus mitigation strategies in variational quantum eigensolver.” *arXiv preprint*, 2025. [arXiv:2512.11171](#)